

Introduction to C++

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What is C++?

- C++ is a cross-platform language that can be used to create high-performance applications.
- C++ was developed by Bjarne Stroustrup, as an extension to the C language.
- C++ gives programmers a high level of control over system resources and memory.
- The language was updated 5 major times in 2011, 2014, 2017, 2020, and 2023 to C++11, C++14, C++17, C++20, and C++23.



Why Use C++

- C++ is one of the world's most popular programming languages. It can be found in today's operating systems, Graphical User Interfaces, and embedded systems.
- C++ is an object-oriented programming language which gives a clear structure to programs and allows code to be reused, lowering development costs.
- C++ is portable and can be used to develop applications that can be adapted to multiple platforms.
- C++ is fun and easy to learn!
- As C++ is close to [C](#), [C#](#) and [Java](#), it makes it easy for programmers to switch to C++ or vice versa.



Difference between C and C++



- C++ was developed as an extension of C, and both languages have almost the same syntax.
- The main difference between C and C++ is that C++ supports classes and objects, while C does not.



Example:

```
#include <iostream>

int main() {
    std::cout << "Hello World!";
    return 0;
}
```

- Line 1: `#include <iostream>` is a header file library that lets us work with input and output objects, such as `cout` (used in line 5). Header files add functionality to C++ programs.
- Line 2: `using namespace std` means that we can use names for objects and variables from the standard library.
- Line 3: A blank line. C++ ignores white space. But we use it to make the code more readable.
- Line 4: Another thing that always appear in a C++ program is `int main()`. This is called a function. Any code inside its curly brackets `{}` will be executed.



Example:

```
#include <iostream>

int main() {
    std::cout << "Hello World!";
    return 0;
}
```

- Line 5: cout (pronounced "see-out") is an object used together with the insertion operator (<<) to output/print text. In our example, it will output "Hello World!".

Note: C++ is case-sensitive: "cout" and "Cout" has different meaning.

Note: Every C++ statement ends with a semicolon ;.

Note: The body of int main() could also been written as:

→int main () { cout << "Hello World! "; return 0; }

Remember: The compiler ignores white spaces. However, multiple lines makes the code more readable.

- Line 6: return 0; ends the main function.
- Line 7: Do not forget to add the closing curly bracket } to actually end the main function.



C++ Statements

- A computer program is a list of "instructions" to be "executed" by a computer.
- In a programming language, these programming instructions are called statements.
- The following statement "instructs" the compiler to print the text "Hello World" to the screen:

```
cout << "Hello World!";
```

It is important that you end the statement with a semicolon ;



Many Statements:

- Most C++ programs contain many statements.
- The statements are executed, one by one, in the same order as they are written:

```
cout << "Hello World!";  
cout << "Have a good day!";  
return 0;
```

- Example explained
- From the example above, we have three statements:
 1. `cout << "Hello World!";`
 2. `cout << "Have a good day!";`
 3. `return 0;`



References

- <https://www.w3schools.com/cpp/default.asp>

